

Basic Python Programming

Introduction to Python

Course: Introduction to Programming Lesson: Basic Python Programming Level: Beginner

Learning Outcomes

By the end of this lesson, students will be able to:

- Explain what Python is.
- Install and open Python.
- Write their first Python program.
- Use variables.
- Work with different data types.
- Accept user input.
- Perform calculations.
- Make decisions using if statements.
- Repeat actions using loops.

Lesson 1: What is Python?

Definition

Python is a high-level programming language used to build:

- Websites
- Mobile applications
- Desktop software
- Artificial Intelligence (AI)
- Data Analysis
- Cybersecurity tools
- Automation scripts

Why Learn Python?

- Easy to read
- Easy to learn
- Free
- Popular worldwide
- Used by Google, Microsoft, Netflix and NASA

Lesson 2: Installing Python

Download Python from: <https://www.python.org>

Download and install:

- Python 3.x
- Tick "Add Python to PATH"
- Click Install

Lesson 3: First Python Program

Code

```
print("Hello World")
```

Output

```
Hello World
```

Explain: print() displays information on the screen.

Lesson 4: Comments

```
# This is a comment  
print("Welcome")
```

Comments are ignored by Python. Shortcut in VS Code: Ctrl + /

Lesson 5: Variables

```
name = "John"  
age = 20  
height = 1.75  
print(name)  
print(age)  
print(height)
```

Output

```
John 20 1.75
```

Lesson 6: Data Types

Type	Example
String	"John"
Integer	25
Float	10.5
Boolean	True

Example: name = "Peter" ...

Lesson 7: User Input

```
name = input("Enter your name: ")
```

```
print("Hello", name)
```

Example

```
Enter your name: Alice
```

```
Hello Alice
```

Lesson 8: Numbers

```
num1 = 20
```

```
num2 = 10
```

```
print(num1 + num2)
```

```
print(num1 - num2)
```

```
print(num1 * num2)
```

```
print(num1 / num2)
```

Output

```
30
```

```
10
```

```
200
```

```
2.0
```

Lesson 9: Calculator Example

```
number1 = int(input("Enter first number: "))
```

```
number2 = int(input("Enter second number: "))
```

```
total = number1 + number2
```

```
print("Answer =", total)
```

Lesson 10: Decision Making

```
age = int(input("Enter age: "))
```

```
if age >= 18:
    print("Adult")
else:
    print("Minor")
```

Explain:

Python uses indentation instead of braces {}.

Lesson 11: Comparison Operators

Operator	Meaning
==	Equal
!=	Not Equal
>	Greater Than
<	Less Than
>=	Greater or Equal
<=	Less or Equal

Example

```
number = 15
```

```
if number > 10:
    print("Greater than 10")
```

Lesson 12: Loops

For Loop

```
for i in range(5):
    print(i)
```

Output

```
0
1
2
3
4
```

While Loop

```
count = 1
```

```
while count <= 5:
    print(count)
    count += 1
```

Lesson 13: Lists

```
fruits = ["Apple", "Orange", "Banana"]
```

```
print(fruits)
```

```
print(fruits[0])
```

Output

Apple

Orange

Banana

Apple

Lesson 14: Functions

```
def greet():
```

```
    print("Welcome Students")
```

```
greet()
```

Output

Welcome Students

Lesson 15: Function with Parameters

```
def greet(name):
```

```
    print("Hello", name)
```

```
greet("John")
```

Lesson 16: Simple Project

Student Marks

```
name = input("Student Name: ")
```

```
mark = int(input("Enter mark: "))
```

```
if mark >= 50:
```

```
    print(name, "Passed")
```

```
else:
```

```
    print(name, "Failed")
```

Lesson 17: Mini Project

Multiplication Table

```
number = int(input("Enter number: "))
```

```
for i in range(1,11):  
    print(number,"x",i,"=",number*i)  
Example  
5 x 1 = 5  
5 x 2 = 10  
...  
5 x 10 = 50
```

Common Errors

Missing quotation marks

```
print(Hello)  
Correct  
print("Hello")
```

Incorrect indentation

Wrong

```
if age > 18:  
print("Adult")
```

Correct

```
if age > 18:  
    print("Adult")
```

Using = instead of ==

Wrong

```
if age = 18:
```

Correct

```
if age == 18:
```

Practical Exercise

create a program that:

1. Asks for their name.
2. Asks for their age.
3. Displays a welcome message.
4. Determines whether they are an adult or a minor.
5. Displays numbers from 1 to 10 using a loop.

Example solution:

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))

print("Welcome", name)

if age >= 18:
    print("You are an adult.")
else:
    print("You are a minor.")

for i in range(1, 11):
    print(i)
```

Summary

By the end of this lesson, you should understand:

- What Python is and where it is used.
- How to write and run simple Python programs.
- Variables and common data types.
- User input and output.
- Arithmetic operations.
- Decision making with if statements.
- Repetition using for and while loops.
- Basic functions and lists.